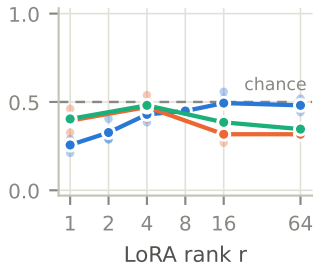
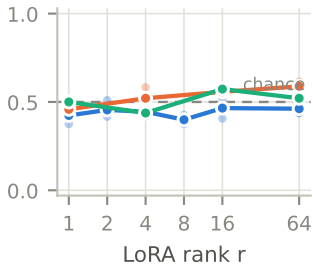


Optimizer ablation, layer 18 down_proj (AdamW lr 2e-4; Muon, AdaHessian at calibrated LR)

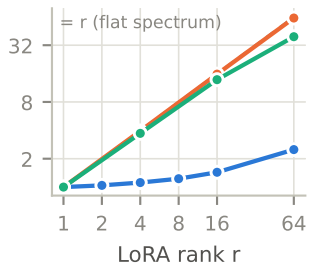
Unseen: A/B accuracy



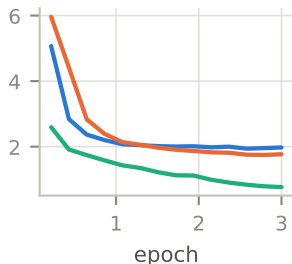
Seen: A/B accuracy



Effective rank of dW



Training loss (r = 16)



—●— AdamW —●— Muon —●— AdaHessian